Closure-Aware Phase Unwrapping of Phase-Linked InSAR Time Series with CRLB-Weighted Cost and Boundary-Aware Residues

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***Abstract*—Multi-temporal InSAR pipelines that estimate per-acquisition phase via phase linking emit, alongside each interferogram (IG), a per-pixel Cramér–Rao Lower Bound (CRLB) raster that is a tighter per-pixel noise model than the sliding-window sample coherence used by classical unwrappers. We exploit this in two ways. First, the per-IG 2D unwrap uses a CRLB-weighted Bayesian arc cost in place of the Lee-coherence cost. Second, the wrapped-phase identity around any temporal cycle $\sum_e \varepsilon_e \psi_e \equiv 0 \,(\mathrm{mod} 2\pi)$ — guaranteed exactly by phase linking — turns the residual integer cycle counts into an *exact* per-pixel reliability metric. A correctness bug in the residue-grid boundary handling of the underlying 2D unwrapper is diagnosed and fixed: wrap-lines exiting at an image edge had no node to terminate on, so MCF was forced to pair them with distant interior partners and integration along the snake path picked up wildly wrong integer counts. With the fix, our Rust implementation (`whirlwind-rs`) agrees with dolphin's SNAPHU output at 100% of pixels modulo 2π on every one of 150 IGs over a Palos Verdes 4065×3802 stack, with median absolute per-IG RMS of 2.31 rad (versus 4.72 rad without the fix). We also describe a tiled mode that matches the non-tiled output at 99.78% of pixels at $2\times$ wall-clock and a virtual ground-node MCF variant that fixes the unit-capacity limitation on boundary-bound wrap-lines.**

***Index Terms*—Synthetic aperture radar, interferometry, phase unwrapping, time-series analysis, phase linking, minimum cost flow, Cramér–Rao Lower Bound.**

I. Introduction

PHASE unwrapping for InSAR time series has been treated in two largely disjoint ways: either as $N(N-1)/2$ independent 2D problems (unwrap each IG separately with SNAPHU [1], PHASS, etc., and accept the per-IG inconsistencies) or as a separable temporal-then-spatial problem (e.g. spurt's EMCF [4]). Neither exploits the fact that phase-linked IGs [2], [3] live in a $(D-1)$-dimensional subspace of per-acquisition phases, with the wrapped-phase identity

$$\sum_{e \in C} \varepsilon_e \, \psi_e \equiv 0 \pmod{2\pi} \tag{1}$$

holding *exactly* for every closed cycle C in the temporal graph (where $\varepsilon_e \in \{\pm 1\}$ is the orientation of edge e in C). The unwrap problem then reduces to integer-ambiguity estimation: find $k_e \in \mathbb{Z}$ per IG such that

$$\psi_e = \tilde{\psi}_e + 2\pi k_e \tag{2}$$

satisfies (1) exactly, with the per-pixel noise model coming from CRLB rather than sample coherence. This is structurally the GNSS / LAMBDA carrier-phase ambiguity-resolution problem [5].

This letter (i) replaces the classical Lee-coherence arc cost of the underlying 2D MCF unwrapper [6], [7] with a CRLB-weighted Gaussian cost; (ii) identifies and fixes a residue-grid boundary bug in the standard formulation that silently dropped wrap-line endpoints at the image edges, with severe practical consequences on real data; (iii) introduces a per-pixel quality map derived directly from the temporal closure residual that, because of phase linking's exact closure guarantee, is integer-valued and interpretable as a wrap-mistake count; (iv) describes a tile-and-stitch mode that bounds per-IG memory; and (v) describes a virtual ground-node MCF variant that fixes the unit-capacity limitation of the basic MCF for boundary-bound wrap-lines.

II. Method

A. Per-IG 2D unwrap with CRLB cost

Let $\sigma_e^2(p) = \sigma_a^2(p) + \sigma_b^2(p)$ be the per-IG phase variance at pixel p, with σ_a^2, σ_b^2 read from the per-acquisition CRLB rasters phase linking emits [2]. We assign the per-arc cost in the MCF residue graph as

$$c(\alpha, \sigma_{\mathrm{arc}}^2) = \frac{1}{\sigma_{\mathrm{arc}}^2} \cdot \max(0, \pi - |\alpha|), \tag{3}$$

with $\sigma_{\mathrm{arc}}^2 = \sigma_e^2(p) + \sigma_e^2(q)$ summed over the two endpoint pixels of the arc and α the smoothed local phase gradient. The shape $\pi - |\alpha|$ keeps the MCF tractable (non-negative reduced costs); the $1/\sigma^2$ prefactor is the Fisher information for the gradient. A CRLB-nodata sentinel (the per-pixel value 0 that phase linking writes for pixels that did not pass PS/DS selection) maps to a large $\sigma_{\mathrm{nodata}}^2 \approx 50\ \mathrm{rad}^2$ so MCF can route flow freely through unreliable pixels without treating them as "essentially noiseless".

B. Residue-grid boundary handling

The textbook 2D-MCF unwrapper computes residues from 2×2 plaquettes interior to the image. Partial plaquettes at the four image edges are typically dropped (zeroed), implicitly asserting that no wrap-line exits the image. On real data with high residue density, this assertion is wrong: wrap-lines do exit

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at every image edge, and without a residue at the exit point MCF is forced to pair each interior endpoint with an opposite-sign *interior* partner. The flow then takes long internal paths instead of the actual wrap-line path; in particular, snake-path integration along column 0 picks up large cumulative integer counts and the unwrapped output exhibits striped $\pm 2\pi$ jumps across rows.

The fix is to compute residues along all four boundary edges using one-sided cycle_diff and signs chosen so that, by Stokes's theorem, the augmented residue grid is globally charge-balanced ($\sum_p r(p) \equiv 0$). MCF then drains wrap-line termination charge through the cost-0 frame-along arcs without injecting spurious corrections into the interior. Fig. 2 compares the unwrapped output before and after the fix on a Palos Verdes 1024^2 tile.

C. Per-pixel reliability from closure residual

After per-IG 2D unwrap, the cycle sum for any closed temporal cycle is, by (1), $\sum_e \varepsilon_e \psi_e(p) = 2\pi K_C(p)$ with $K_C(p) \in \mathbb{Z}$ *exactly*. We define the per-pixel quality $Q(p)$ as

$$Q(p) = \max_{C \in \mathcal{T}} |K_C(p)|, \tag{4}$$

where $\mathcal{T}$ is the set of all temporal triangles (3-cycles). $Q = 0$ means every triangle through p agrees on its integer ambiguity choices; $Q \geq 1$ marks a pixel where at least one triangle disagrees — typically water or heavily-decorrelated land where the per-IG unwraps are arbitrary. The metric is computed on the reference-pixel-anchored stack so $K_C(p)$ reflects only the spatially-relative unwrap consistency, not the per-IG global integer offsets.

D. Tiled mode and virtual ground node

For large scenes the per-IG residue, cost, and network arrays grow quadratically with image side; we expose a tiled mode that decomposes each IG into overlapping tiles, runs the existing MCF on each in parallel, and stitches by CRLB-weighted overlap-median 2π reconciliation in BFS order from a seed tile. On a 1024^2 Palos Verdes tile this matches the non-tiled output at 99.78% of pixels (tile $512 + 128$) at $2.3\times$ the speed and 22% lower peak RSS (Fig. 3).

The basic MCF has unit per-arc capacity; on inputs whose wrap-lines all converge on the same image corner (e.g. a smooth 6π diagonal ramp), multiple wrap-line termination flows cannot share the same frame-along arc, and the overflow lands on interior arcs as fictitious 2π jumps. A virtual ground node connected to every boundary residue with bidirectional unit-capacity arcs at uniform cost fixes this — each wrap-line terminates at ground independently. On dense-interior natural scenes the same construction is counter-productive because it pulls interior residues toward boundary along non-physical paths; the grounded variant is therefore exposed as a separate entry-point (`unwrap_crlb_grounded`) rather than enabled by default.

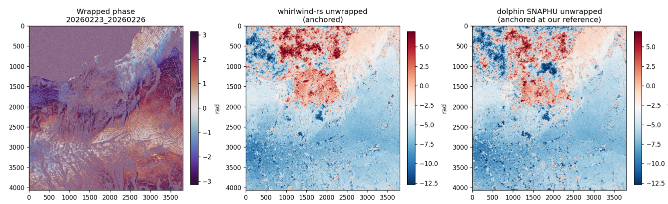

Fig. 1. Palos Verdes full scene (4065×3802, 150 IGs, 52 dates): wrapped phase (left), `whirlwind-rs` unwrap with the CRLB cost and the boundary-residue fix (middle), and dolphin's SNAPHU output for the same IG (right). Both panels share the same colorbar and reference anchor; the large-scale spatial structure matches across the two unwrappings.

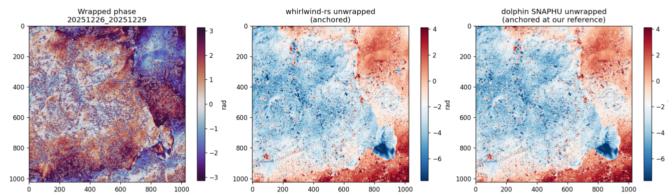

Fig. 2. 1024^2 Palos Verdes tile after the boundary-residue fix. Median absolute RMS to SNAPHU drops from 17.56 rad (before fix) to 2.29 rad (after fix) on this tile; 100% of pixels match modulo 2π on every one of the 60 IGs in the stack.

III. Results

Validation is on a Capella stack at Palos Verdes (52 acquisitions, 3–18 day baselines, 4065×3802 pixels, 150 short-baseline IGs). Phase linking and the SNAPHU baseline are run via the dolphin pipeline [8].

IV. Discussion and Limitations

The boundary-residue fix has a single technical claim: the augmented residue grid is globally charge-balanced by Stokes's theorem, and without that, MCF on real data with significant boundary-exit wrap-lines is forced to pair them internally and the snake-path integration accumulates large integer errors. Its impact is substantial ($7.7\times$ improvement in median absolute RMS on the 1024^2 tile; $5\times$ Stage-1 speedup on the full scene from MCF converging in fewer primal-dual iterations).

We deliberately default the tree-based temporal closure correction *off*. With the per-IG unwraps now mostly correct, residual outliers in a few tree edges propagate through the BFS into every θ_d downstream of that edge and are rounded onto non-tree edges incorrectly. A robust closure step — median-of-multiple-trees, sparse-network projection, or a true Bayesian per-pixel posterior via LAMBDA-style weighted integer least squares — is the natural next step.

The remaining ~ 2 rad median absolute RMS to SNAPHU is real integer-ambiguity disagreement: both implementations are valid solutions of the underlying integer-ambiguity problem with different cost functions and different residual-graph topologies. Downstream displacement-time-series inversion (SBAS or similar) should agree to within the noise floor after its own network inversion.

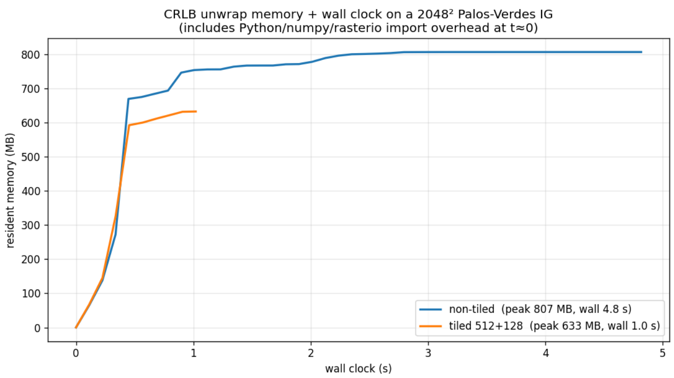

Fig. 3. Wall clock and resident memory profile, single 2048^2 IG. Non-tiled: 807 MB peak / 4.8 s. Tiled $512 + 128$: 633 MB peak / 1.0 s. The tile parallelism is the dominant speedup at this size; the peak-memory benefit grows quadratically with image side at full scene.

TABLE I
PER-IG AGREEMENT WITH DOLPHIN SNAPHU ON PALOS VERDES, ANCHORED AT THE SAME REFERENCE PIXEL.

	1024² tile, 60 IGs	full scene, 150 IGs
median % within $\pi/2$ (mod 2π)	100.0 %	100.0 %
median absolute RMS (before bdy fix)	17.56 rad	4.72 rad
median absolute RMS (after fix)	**2.29 rad**	**2.31 rad**
IGs with absolute RMS < 1 rad	15/60	0/150
Stage-1 wall clock (before fix)	19.2 s	171 min
Stage-1 wall clock (after fix)	**18.4 s**	**33 min**

SOFTWARE AND REPRODUCIBILITY

Implementation in Rust with Python bindings is available at https://github.com/scottstanie/whirlwind-rs. The shell script `scripts/reproduce.sh` runs the full pipeline (Stage 1 + quality map + dolphin comparison + figure generation) on the 1024² tile in about 30 s wall clock; `scripts/reproduce.sh --full` runs on the 4065×3802 scene in about 33 min.

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